The Theory of Value-Based Accounting

A Stochastic Decomposition of the Accounting Identity into Value-Trajectory Classes: Mathematics, Conservation Laws, and Multidisciplinary Correspondences

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Abstract

The accounting identity $A = L + E$ constrains the *level* of the balance sheet but is silent about its *dynamics*. This paper decomposes each side of the identity into three value-trajectory classes—increasing, constant, decreasing—yielding nine stochastic components ($A_I, A_C, A_D, L_I, L_C, L_D, E_I, E_C, E_D$) governed by five axioms. We prove a canonical and unique attribution of equity (Theorem 7), a pathwise Flow Conservation Law (Theorem 8), a Drift Balance theorem with an associated scalar *value momentum* $M = C - D$ (Theorem 9), an erosion bound with an explicit expected insolvency time, and a Covariance Singularity theorem: the value state lives in an 8-dimensional hyperplane and, under the canonical model, its stochastic rank is at most 6 (at most 4 under strict constancy). The identity is shown to be gauge-invariant under measure and numéraire changes while the trajectory classification is gauge-covariant. The framework is then connected to physics (Noether conservation and gauge symmetry), chemistry (stoichiometry and reaction networks), biology (anabolism and catabolism), psychology (mental accounting and loss aversion), psychiatry (clinical staging of erosion), philosophy and logic (analytic identity versus synthetic drifts, soundness and completeness), religion (Jubilee, usury limits, and zakat as value-flow institutions), political science (Goodhart's law), diplomacy and international relations (verification regimes), welfare economics (social value momentum), warfare economics (attrition and logistics), statistics (state-space estimation and rank tests), and computer science, data science, and AI/ML (algorithms, complexity, and ledger-authenticity classification).

The paper ends with "The End".

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1 Introduction

The identity
$$A = L + E \tag{1}$$
with assets A, liabilities L, and shareholders' equity E is the oldest conservation law in commerce. It is also the most inert: it constrains a snapshot and says nothing about motion. A balance sheet can satisfy (1) exactly while describing an enterprise that is compounding, stagnating, or disintegrating.

This paper asks a dynamics question:

What stochastic structure must the components of the balance sheet obey for the identity to hold at every instant?

We answer by splitting each of A, L, E into three value-trajectory classes and modeling the nine resulting components as adapted stochastic processes. The answer is that the nine-tuple is heavily—but not completely—constrained. It lives on an 8-dimensional hyperplane, its dynamics are generated by four primitive value flows, and its covariance structure is necessarily singular.

The result continues the program of two preceding papers. The first established that logistic regression learns equivalence classes of representations under affine gauge symmetries [1]; the second established that authenticity detection requires leaving a model's invariance class [2]. Here the accounting identity plays the role of the invariant, and the value-trajectory classification plays the role of a frame-dependent observable. A forged ledger satisfies the invariant trivially; we show it struggles to satisfy the motion.

2 The Probability Space and the Ninefold Decomposition

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions [9]. A firm holds assets $i = 1, \dots, n$ with value processes $V_i(t) \geq 0$ and owes liabilities $j = 1, \dots, m$ with obligation processes $B_j(t) \geq 0$, all adapted semimartingales:

$$A(t) = \sum_{i=1}^{n} V_i(t), \qquad L(t) = \sum_{j=1}^{m} B_j(t), \qquad E(t) = A(t) - L(t). \tag{2}$$

2.1 Trajectory classes and value buckets

Definition 1 (Trajectory Class)**.** Fix a materiality threshold $\varepsilon \geq 0$ and a reporting period $[0, T]$ on which classifications are frozen. An asset with expected value drift $\mu_i(t) = \frac{d}{dt}\mathbb{E}[V_i(t)]$ is *increasing* if $\mu_i(t) > \varepsilon$, *constant* if $|\mu_i(t)| \leq \varepsilon$, and *decreasing* if $\mu_i(t) < -\varepsilon$. The same trichotomy applies to liabilities via their drifts $\nu_j(t)$.

Definition 2 (Value Buckets)**.** Let $\mathsf{A}_I, \mathsf{A}_C, \mathsf{A}_D \subset \{1, \dots, n\}$ and $\mathsf{L}_I, \mathsf{L}_C, \mathsf{L}_D \subset \{1, \dots, m\}$ be the index sets of Definition 1. Define

$$A_I = \sum_{i\in\mathsf{A}_I} V_i, \quad A_C = \sum_{i\in\mathsf{A}_C} V_i, \quad A_D = \sum_{i\in\mathsf{A}_D} V_i, \tag{3}$$

and identically for L_I, L_C, L_D. Equity is decomposed in Section 4.

Definition 3 (Value-State Vector)**.** The *value-state vector* is

$$X(t) = \begin{pmatrix} A_I & A_C & A_D & L_I & L_C & L_D & E_I & E_C & E_D \end{pmatrix}^\top \in \mathbb{R}^9. \tag{4}$$

Define the *balance functional*

$$b^\top x \;=\; (x_1 + x_2 + x_3) - (x_4 + x_5 + x_6 + x_7 + x_8 + x_9), \tag{5}$$

so that $b = (1, 1, 1, -1, -1, -1, -1, -1, -1)^\top$. The value state lives on the *value hyperplane*

$$\mathcal{V} \;=\; \{\, x \in \mathbb{R}^9 : b^\top x = 0 \,\}, \tag{6}$$

an affine slice of dimension 8.

3 Axioms of Value-Based Accounting

Axiom 1 (Trichotomy)**.** Each class triple partitions its register:

$$A = A_I + A_C + A_D, \qquad L = L_I + L_C + L_D, \qquad E = E_I + E_C + E_D \tag{7}$$

almost surely, for all t.

Axiom 2 (Balance)**.** $b^\top X(t) = 0$ almost surely for all t.

Axiom 3 (Trajectory)**.** The expected value velocities $\nu_X(t) = \frac{d}{dt}\mathbb{E}[X(t)]$ have fixed signs:

$$\begin{aligned} &\nu_{A_I} > 0, \quad \nu_{A_C} = 0, \quad \nu_{A_D} < 0, \\ &\nu_{L_I} > 0, \quad \nu_{L_C} = 0, \quad \nu_{L_D} < 0, \\ &\nu_{E_I} > 0, \quad \nu_{E_C} = 0, \quad \nu_{E_D} < 0. \end{aligned} \tag{8}$$

Remark 4 (Weak versus strict constancy). "Constant in value" admits two readings: (i) *weak*—zero drift, so the class is a martingale: expected value constant, volatility permitted; appropriate for mark-to-market portfolios; (ii) *strict*—pathwise constant, $A_C(t) = A_C(0)$ a.s.: appropriate for cash and par instruments. Pathwise monotonicity would force deterministic dynamics, so the drift-sign reading of Axiom 3 is the only non-degenerate stochastic interpretation. *Increasing in value* therefore means *increasing in expected value.*

Axiom 4 (Support and Solvency)**.** $A_* \geq 0$ and $L_* \geq 0$ componentwise; the equity buckets may pass through zero and become negative, recording erosion of prior tranches.

Axiom 5 (Attribution / Purity)**.** Equity value flows are generated only by asset and liability value flows: equity-increasing flow is generated by asset appreciation and liability amortization; equity-decreasing flow by asset depreciation and liability accretion; every primary flow is attributed exactly once; martingale noise of the constant classes is absorbed by E_C.

4 The Canonical Stochastic Model

4.1 Primitive flows and stochastic differential equations

Let $W_1, \ldots, W_6$ be (possibly correlated) standard Brownian motions and let the four *primitive rates* be strictly positive:

$$\alpha \text{ (appreciation)}, \quad \beta \text{ (depreciation)}, \quad \gamma \text{ (accretion)}, \quad \delta \text{ (amortization)}. \tag{9}$$

Model 1 (Canonical System)**.**

$$dA_I = +\alpha A_I\,dt + \sigma_1 A_I\,dW_1, \qquad dA_C = \sigma_2 A_C\,dW_2, \qquad dA_D = -\beta A_D\,dt + \sigma_3 A_D\,dW_3, \tag{10}$$
$$dL_I = +\gamma L_I\,dt + \sigma_4 L_I\,dW_4, \qquad dL_C = \sigma_5 L_C\,dW_5, \qquad dL_D = -\delta L_D\,dt + \sigma_6 L_D\,dW_6, \tag{11}$$
$$dE_I = dA_I - dL_D, \qquad dE_C = dA_C - dL_C, \qquad dE_D = dA_D - dL_I, \tag{12}$$

with initial conditions satisfying $E_I(0) + E_C(0) + E_D(0) = A(0) - L(0)$.

The equity rows are the *canonical attribution*: equity is long appreciating assets and short amortizing liabilities (hence E_I grows), and long depreciating assets and short accreting liabilities (hence E_D falls).

4.2 Solutions and well-posedness

Proposition 5 (Telescoping Solutions)**.** *The equity buckets are running tallies of primary flows:*

$$E_I(t) = E_I(0) + \big[A_I(t) - A_I(0)\big] + \big[L_D(0) - L_D(t)\big], \tag{13}$$
$$E_D(t) = E_D(0) + \big[A_D(t) - A_D(0)\big] + \big[L_I(0) - L_I(t)\big], \tag{14}$$
$$E_C(t) = E_C(0) + \big[A_C(t) - A_C(0)\big] - \big[L_C(t) - L_C(0)\big]. \tag{15}$$

Consequently $E_I + E_C + E_D = E$ for all t.

Proof. Integrate (12) row by row. Summing the three integrated equations and using $E(0) = A(0) - L(0)$ gives $E_I(t) + E_C(t) + E_D(t) = E(0) + [A(t) - A(0)] - [L(t) - L(0)] = E(t)$. □

Proposition 6 (Well-Posedness)**.** *The system* (12) *is linear with globally Lipschitz coefficients; strong solutions exist and are unique [9]; the geometric Brownian buckets A_*, L_* remain strictly positive; and Axioms 1–5 all hold. Under strict constancy set $\sigma_2 = \sigma_5 = 0$.*

5 Core Theorems

5.1 Uniqueness of the equity attribution

Theorem 7 (Canonical Attribution)**.** *Given asset and liability dynamics and initial conditions, there is exactly one equity decomposition satisfying Axioms 1–5, namely the attribution of Model 1.*

Proof. Purity (Axiom 5) restricts the drivers of dE_I to the span of $\{dA_I, -dL_D\}$ and of dE_D to the span of $\{dA_D, -dL_I\}$. Since the total equity flow must satisfy $dE_I + dE_D = dA_I + dA_D - dL_I - dL_D$, and each primary flow is attributed exactly once, the coefficients of dA_I and $-dL_D$ in dE_I and of dA_D and $-dL_I$ in dE_D are forced to equal 1. The constant-class identity $dE_C = dA_C - dL_C$ then follows from Axiom 1, and initial conditions are fixed by Proposition 5. □

5.2 Flow conservation: the First Law

Theorem 8 (Flow Conservation Law)**.** *Under the canonical attribution, pathwise for all t:*

$$dA_I + dA_C + dA_D \;=\; dL_I + dL_C + dL_D + dE_I + dE_C + dE_D, \tag{16}$$

and, since $dE_C = dA_C - dL_C$ exactly,

$$\boxed{dA_I + dA_D \;=\; dL_I + dL_D + dE_I + dE_D} \qquad a.s. \tag{17}$$

Proof. Differentiate Axiom 2 to obtain (16). Substituting $dE_C = dA_C - dL_C$ cancels the constant classes and yields (17). □

Equation (17) is a conservation law in the sense of Noether [16]: the symmetry of the balance sheet under time translations of its transactions induces an invariant differential charge $b^\top dX = 0$.

5.3 Drift balance and value momentum

Theorem 9 (Drift Balance)**.** *Define expected value velocities $\nu_X(t) = \frac{d}{dt}\mathbb{E}[X(t)]$. Then*

$$\nu_{A_I} + \nu_{A_D} \;=\; \nu_{L_I} + \nu_{L_D} + \nu_{E_I} + \nu_{E_D}. \tag{18}$$

Proof. Take expectations of (17); the constant classes contribute zero by Axiom 3. □

Definition 10 (Value Momentum)**.** Let

$$M(t) \;:=\; \nu_{E_I} + \nu_{E_D} \;=\; \big(\nu_{A_I} + \nu_{A_D}\big) - \big(\nu_{L_I} + \nu_{L_D}\big). \tag{19}$$

Writing the gross creation and destruction rates

$$C \;=\; \nu_{A_I} + |\nu_{L_D}|, \qquad D \;=\; |\nu_{A_D}| + \nu_{L_I}, \tag{20}$$

we obtain $M = C - D$. Positive M means the firm creates value; the entire balance-sheet dynamics reduce to four primitive flows and one scalar.

5.4 Erosion and insolvency

Proposition 11 (Erosion Bound)**.** *Suppose the velocity vector is constant on $[0,T]$ (arithmetic approximation). Then*

$$\mathbb{E}[E(t)] = E(0) + Mt. \tag{21}$$

If $M < 0$, expected insolvency arrives at

$$t^* \;=\; \frac{E(0)}{-M}. \tag{22}$$

5.5 Covariance singularity: second-order balance

Under Model 1 the state obeys

$$dX_t \;=\; \mu(X_t)\,dt \;+\; \Lambda(X_t)\,dW_t, \tag{23}$$

where $dW_t = (dW_1, \ldots, dW_6)^\top$ and

$$\Lambda(X) \;=\; \begin{pmatrix} \sigma_1 A_I & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_2 A_C & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_3 A_D & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_4 L_I & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_5 L_C & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_6 L_D \\ \sigma_1 A_I & 0 & 0 & 0 & 0 & -\sigma_6 L_D \\ 0 & \sigma_2 A_C & 0 & 0 & -\sigma_5 L_C & 0 \\ 0 & 0 & \sigma_3 A_D & -\sigma_4 L_I & 0 & 0 \end{pmatrix}. \tag{24}$$

Lemma 12 (Balance of the Diffusion)**.** $b^\top \Lambda(X) = 0^\top$ *for all admissible X.*

Proof. Each column of Λ carries one primitive flow with the same coefficient in exactly one asset or liability bucket and the opposite coefficient in exactly one equity bucket; the six cancellations $\sigma_1 A_I - \sigma_1 A_I$, $\sigma_2 A_C - \sigma_2 A_C$, $\sigma_3 A_D - \sigma_3 A_D$, $-\sigma_4 L_I + \sigma_4 L_I$, $-\sigma_5 L_C + \sigma_5 L_C$, and $-\sigma_6 L_D + \sigma_6 L_D$ are read off column by column. □

Lemma 13 (Factor Independence)**.** *If all six buckets are strictly positive and all $\sigma_k > 0$, then* $\operatorname{rank}\Lambda = 6$*. Under strict constancy ($\sigma_2 = \sigma_5 = 0$),* $\operatorname{rank}\Lambda \le 4$*.*

Proof. Let $\sum_{k=1}^{6} c_k \Lambda_{\cdot k} = 0$. Reading rows $A_I, A_C, A_D, L_I, L_C, L_D$ successively forces $c_1 = \cdots = c_6 = 0$, proving independence. Setting $\sigma_2 = \sigma_5 = 0$ deletes two independent columns. □

Theorem 14 (Covariance Singularity)**.** *For any nine-tuple satisfying Axioms 1 and 2,*

$$\Sigma_X\, b = 0, \qquad b^\top \Sigma_X\, b = 0, \qquad \operatorname{rank}(\Sigma_X) \le 8. \tag{25}$$

Under Model 1 with independent Brownian motions, $\Sigma_X = \Lambda\Lambda^\top$, hence

$$\operatorname{rank}(\Sigma_X) \le 6, \tag{26}$$

and at most 4 under strict constancy.

Proof. Since $b^\top X = 0$ a.s., $\operatorname{Var}(b^\top X) = b^\top \Sigma_X b = 0$ and $\operatorname{Cov}(X, b^\top X) = \Sigma_X b = 0$. The rank bounds follow from Lemmas 12 and 13 because the column space of $\Sigma_X = \Lambda\Lambda^\top$ is contained in the column space of Λ. □

Corollary 15 (Factor Representation)**.** *The whole balance sheet is spanned by four primitive value flows: appreciation, depreciation, accretion, amortization. Equity is bookkeeping for these flows.*

Example 16 (Variance of equity)**.** By Proposition 5, $E(T) = \text{const} + A_I(T) + A_D(T) - L_I(T) - L_D(T)$, so $\operatorname{Var}(E(T)) = v^\top \Sigma^{(4)} v$ with $v = (1, 1, -1, -1)^\top$ over the four flowing buckets. For independent geometric Brownian buckets,

$$\operatorname{Var}\left[V(T)\right] = V(0)^2\, e^{2\mu T}\left(e^{\sigma^2 T} - 1\right), \tag{27}$$

and $\operatorname{Var}(E(T))$ is the corresponding sum of four such terms [8].

5.6 Gauge invariance and gauge covariance

Theorem 17 (Gauge Dependence of Classification)**.** *Under any equivalent measure change (Girsanov: with market price of risk θ, drifts shift by $-\sigma_k\theta_k$), or under any numéraire change $\tilde{X} = X/N$ for a positive adapted N:*

1. *the identity $A = L + E$ and Theorems 8–14 remain valid;*
2. *the trajectory classification generally changes: buckets migrate between the I, C, and D classes.*

Therefore the balance identity is gauge-invariant *while the trajectory classification is* gauge-covariant*.*

Proof. The conservation law (17) is pathwise and hence measure-independent; under an equivalent measure it persists with probability one. Drift signs, by contrast, are not invariant: the Girsanov density $d\mathbb{Q}/d\mathbb{P}\big|_{\mathcal{F}_t} = \exp\left(\int_0^t \theta^\top dW_s - \frac{1}{2}\int_0^t \|\theta\|^2 ds\right)$ shifts $\mu_k \mapsto \mu_k - \sigma_k\theta_k$ and can cross the thresholds of Definition 1. The numéraire rescaling acts componentwise and likewise reshuffles signs. □

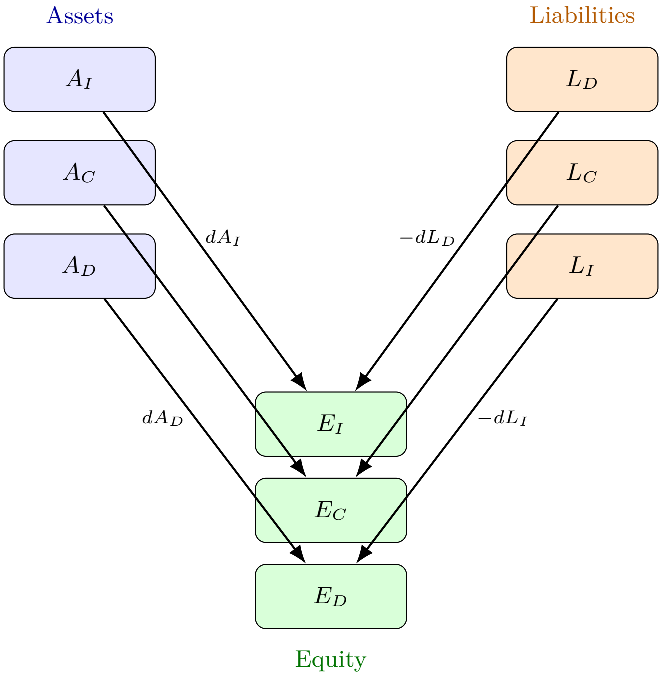

Figure 1: Canonical attribution of value flows.

(a) Equity tranches are running tallies of the four primitive flows (Theorem 7); the constant classes transmit martingale noise into E_C.

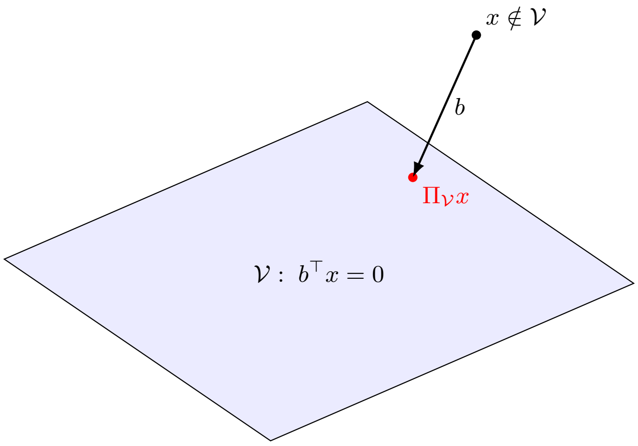

Figure 2: The value state lives on the 8-dimensional hyperplane $\mathcal{V} \subset \mathbb{R}^9$ (Theorem 14)

(a) The balance functional b is the normal direction along which no stochastic variation is possible.

6 Transactions as Conservation Operators

Between reporting dates the organic dynamics of Model 1 hold. Discrete transactions act on the value state through linear operators T satisfying the conservation condition

$$b^\top T = b^\top. \tag{28}$$

These operators form a monoid under composition: double-entry bookkeeping is a conservation law, and the nine buckets are its *charge sectors*.

Table 1: Canonical transactions as conservation operators (eq. 28).

(a) Each row satisfies $\sum \Delta(\text{assets}) = \sum \Delta(\text{liabilities}) + \sum \Delta(\text{equity})$.

Transaction	ΔA_I	ΔA_C	ΔA_D	ΔL_I	ΔL_C	ΔL_D	ΔE_I	ΔE_C	ΔE_D
Revaluation	$+x$	0	0	0	0	0	$+x$	0	0
Impairment	0	0	$-x$	0	0	0	0	0	$-x$
Debt amortization	0	$-x$	0	0	0	$-x$	$+x$	$-x$	0
New borrowing	0	$+x$	0	$+x$	0	0	0	$+x$	$-x$
Distribution	0	$-x$	0	0	0	0	0	0	$-x$
Jubilee (§7.7)	0	0	0	0	0	$-x$	$+x$	0	0

Note that reclassifications are compositions: an impairment is the operator "move x from class I to class D" followed by the destruction flow $A_D \downarrow x,\ E_D \downarrow x$.

7 Interdisciplinary Correspondences

7.1 Physics: conservation laws and gauge symmetry

By Noether's theorem [16], every continuous symmetry of a system's action yields a conserved quantity. Here the symmetry is the redundancy of the balance-sheet representation and the conserved charge is $b^\top X$. The measure and numéraire redundancies of Theorem 17 are genuine gauge freedoms: observables (conservation laws) are invariant, coordinates (trajectory classes) are not. Erosion plays the role of entropy production in dissipative structures [18]: under $M < 0$ the system exports value to its environment irreversibly, and t^* is the analog of a relaxation time.

7.2 Chemistry: stoichiometry and reaction networks

In a chemical reaction network with 9 species and k reactions, conservation laws are exactly the left null space of the stoichiometric matrix $N \in \mathbb{R}^{9\times k}$ [17]. The balance functional b spans the left null space of the transaction stoichiometry of Table 1a: the accounting identity is a stoichiometric constraint, transactions are elementary reactions, and value momentum M is a net reaction flux. Le Chatelier's principle—a perturbation induces compensating flows—reappears as mean reversion between buckets after shocks.

7.3 Biology: metabolism and homeostasis

Gross creation C is anabolism; gross destruction D is catabolism; $M = C - D$ is net growth. Weak-constant classes implement homeostasis (regulated stability under noise), the cumulative destruction $\int_0^t D_s\, ds$ is an allostatic load, and senescence is the regime of persistent $M < 0$ with t^* as the organismal prognosis.

7.4 Psychology: mental accounting and loss aversion

Investors maintain separate cognitive accounts [15]; the trajectory buckets institutionalize mental accounting. Loss aversion [14] with coefficient $\lambda > 1$ implies that perceived momentum is asymmetric,

$$M^{\text{perc}} = C - \lambda D, \qquad \lambda \approx 2.25, \tag{29}$$

so observers demand $C > \lambda D$ before judging an enterprise healthy. The disposition effect is reclassification aversion: a reluctance to move assets from class I to class D while losses are unrealized.

7.5 Psychiatry: clinical staging of value erosion

The tranches support a staging model analogous to clinical staging:

$$\begin{aligned}
&\mathrm{S}_0: \ M > 0 && \text{(compensated)},\\
&\mathrm{S}_1: \ M < 0,\ E_I > |E_D| && \text{(prodromal erosion)},\\
&\mathrm{S}_2: \ E_I + E_D < 0 && \text{(tranche exhaustion)},\\
&\mathrm{S}_3: \ E < 0 && \text{(insolvency)}.
\end{aligned} \tag{30}$$

A firm that reports an exactly balanced identity while ignoring the sign structure of Axiom 3 exhibits the analog of anosognosia: the invariant is preserved while the deficit is denied.

7.6 Philosophy and logic: analytic and synthetic

Because equity is defined as the residual, the identity (1) is *analytic*: true by definition in every frame [19]. The drift signs of Axiom 3 are *synthetic* and empirical, hence falsifiable in the sense of Popper [20]: a single robust observation of $\nu_{E_I} < 0$ jointly with $\nu_{A_I} > 0$ and $\nu_{L_D} < 0$ refutes the canonical attribution. Induction enters through drift estimation from finite series [21]. As a formal theory, Axioms 1–5 are *consistent* (the canonical model is a model: soundness) and Theorem 7 shows the attribution is *complete*: no admissible equity decomposition lies outside the canonical one.

7.7 Religion: Jubilee, usury, and zakat as value-flow institutions

Religious law encodes value-flow institutions long before double-entry bookkeeping. The Jubilee (Leviticus 25:8–13; Deuteronomy 15:1–6) [27] is a periodic reset operator

$$T_{\mathrm{J}}: \quad L_D \mapsto L_D - x, \qquad E_I \mapsto E_I + x, \qquad b^\top T_{\mathrm{J}} = b^\top, \tag{31}$$

which cancels the amortization tranche and restores the creation tranche. Usury prohibitions impose an institutional cap $\gamma \le \bar{\gamma}$ on the accretion rate, and zakat (Qur'an 9:60) [28] is a transfer operator that preserves $b^\top$ on every consolidated ledger while redistributing between registries.

7.8 Political science: credibility and Goodhart's law

When a scalar metric such as M becomes a policy target it ceases to be a good measure [22]: statistical agencies and firms can optimize the reported level. The theory's defense is structural: monitors should test *constraint sets* (Theorems 8–14), which are jointly expensive to satisfy, rather than levels, which are cheap.

7.9 Diplomacy and international relations: verification regimes

Arms-control verification samples compliance mechanisms, not stockpiles. Similarly, sovereign-account inspection should audit conservation residuals and drift-sign consistency across periods. Levels are cheap to fake (papers I–II [1, 2]); constraints are not. Disclosure credibility therefore hinges on the verifiability of $\{b^\top X_t = 0\} \cap \{\text{sign structure}\} \cap \{\rho \approx 0\}$.

7.10 Welfare economics: social value momentum

Let $\mathcal{F}$ be the population of enterprises with measure μ. A social planner values creation and penalizes destructive erosion through its externalities (unemployment, creditor contagion):

$$W \;=\; \int_{\mathcal{F}} u\big(M_f\big)\, d\mu(f) \;-\; \kappa \int_{\mathcal{F}} D_f \, d\mu(f), \tag{32}$$

with u increasing and concave and $\kappa > 0$ the externality coefficient. Because transfers are conservation operators (eq. 28), redistribution preserves every ledger's identity while moving mass between the creation and erosion tranches [24].

7.11 Warfare economics: attrition and logistics

With $L \equiv 0$ and the force bucket playing E, the Flow Conservation Law specializes to the logistics identity

$$S_t = S_{t-1} + P_t - C_t \qquad \Longleftrightarrow \qquad \Delta S = P - C, \tag{33}$$

and $M_{\text{war}} = P - C$ is the attrition momentum [25]. A blockade is an imposed rise in β and a cut in α; coupled force erosion obeys Lanchester dynamics [23],

$$\frac{dA_I}{dt} = -\beta\, L_I, \qquad \frac{dL_I}{dt} = -\alpha\, A_I, \tag{34}$$

and battlefield deception manipulates the enemy's ledger-reading classifier [26].

Table 2: Multidisciplinary correspondence table.

Field	Concept	VBA counterpart
Physics	Noether charge	conservation law $b^\top dX = 0$
Physics	gauge freedom	measure / numéraire redundancy
Chemistry	stoichiometric balance	accounting identity
Chemistry	reaction	transaction operator
Biology	anabolism / catabolism	C / D flows
Biology	homeostasis	weak-constant classes
Psychology	mental accounting	trajectory buckets
Psychology	loss aversion	$M^{\mathrm{perc}} = C - \lambda D$
Psychiatry	clinical staging	erosion stages S_0–S_3
Philosophy	analytic / synthetic	identity / drift signs
Religion	Jubilee	reset operator T_{J}
Political science	Goodhart's law	level-metric gaming
IR / diplomacy	verification regime	constraint-set auditing
Welfare economics	social welfare	W of eq. (32)
Warfare economics	attrition	$M_{\mathrm{war}} = P - C$

8 Data Science, Algorithms, and AI/ML

8.1 State-space estimation

Disclosures are noisy observations of the latent buckets. With $d = 9$ and step size Δt, the Euler discretization of (23) and the measurement equation are

$$X_t = X_{t-1} + \mu(X_{t-1})\,\Delta t + \Lambda_{t-1}\,\Delta W_t, \tag{35}$$
$$y_t = H\,X_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, R), \tag{36}$$

and the Kalman recursions [10, 11] are

$$\hat{X}_{t|t-1} = \hat{X}_{t-1|t-1} + \mu\big(\hat{X}_{t-1|t-1}\big)\,\Delta t, \tag{37}$$
$$P_{t|t-1} = F\,P_{t-1|t-1}\,F^\top + Q, \tag{38}$$
$$K_t = P_{t|t-1}H^\top\big(H\,P_{t|t-1}H^\top + R\big)^{-1}, \tag{39}$$
$$\hat{X}_{t|t} = \hat{X}_{t|t-1} + K_t\big(y_t - H\hat{X}_{t|t-1}\big), \tag{40}$$
$$P_{t|t} = (I - K_t H)\,P_{t|t-1}. \tag{41}$$

8.2 The rank-deficiency statistic

Let $\lambda_1 \geq \cdots \geq \lambda_9$ be the eigenvalues of the sample covariance $\hat{\Sigma}$ of the nine bucket series. By Theorem 14, authentic ledgers satisfy

$$\hat{\rho} \;=\; \frac{\lambda_7 + \lambda_8 + \lambda_9}{\sum_{i=1}^{9} \lambda_i} \;\approx\; 0 \tag{42}$$

(rank at most 6 canonically, 4 under strict constancy), whereas synthetic ledgers generated with independent buckets exhibit $\hat{\rho}$ bounded away from zero. Significance is assessed by bootstrap calibration of $\hat{\rho}$ under the null of rank deficiency, in the spirit of likelihood-ratio testing [13].

8.3 Algorithms and complexity

Table 3: Algorithm 1: Ledger authenticity pipeline.

Input: ledger series $\{X_t\}_{t=1}^{T}$, institutional rate bounds	
1	Identity test: verify $b^\top X_t = 0$ for all t; halt if violated.
2	Drift estimation: compute $\hat{\nu}$ for each bucket by regression.
3	Sign test: check the six inequalities of Axiom 3.
4	Conservation test: compute residuals $r_t = \Delta A_{I,t} + \Delta A_{D,t} - \Delta L_{I,t} - \Delta L_{D,t} - \Delta E_{I,t} - \Delta E_{D,t}$; verify $\lVert r \rVert \approx 0$ (Theorem 8).
5	Covariance: form $\hat{\Sigma}$; compute $\lambda_1 \geq \cdots \geq \lambda_9$.
6	Rank statistic: compute $\hat{\rho}$ of eq. (42); bootstrap confidence interval.
7	Prior test: compare $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\delta})$ to institutional bounds.
8	**Output:** authentic iff all tests pass; otherwise flag the buckets responsible.

Table 4: Algorithm 2: Kalman estimation of latent buckets from noisy disclosures (eqs. 36–41).

Input: disclosures $\{y_t\}$, model (F, Q, H, R), prior $(\hat{X}_0, P_0)$	
1	Initialize $\hat{X}_{0\mid 0} = \hat{X}_0$, $P_{0\mid 0} = P_0$.
2	**for** $t = 1, \ldots, T$: **predict** via $\hat{X}_{t\mid t-1}, P_{t\mid t-1}$; **update** via K_t to $\hat{X}_{t\mid t}, P_{t\mid t}$.
3	Enforce the constraint $b^\top \hat{X}_{t\mid t} = 0$ by projection onto $\mathcal{V}$.
4	Estimate primitive rates from filtered drifts.
5	**Output:** $\{\hat{X}_{t\mid t}\}$, $\hat{\nu}$, $\hat{M}$, t^* estimate.

Complexity. With dimension $d = 9$ and T observations: covariance estimation costs $O(d^2 T)$; the eigendecomposition $O(d^3)$; each Kalman step $O(d^3)$, hence $O(d^3 T)$ overall—linear in T for fixed dimension. Fake-ledger generation, by contrast, must satisfy the joint test set, which is combinatorially expensive in the number of constraints.

8.4 Machine-learning authenticity classification

Following [2], define the structural feature map

$$\Phi(D) \;=\; \big(\Phi_{\mathrm{id}},\, \Phi_{\mathrm{flow}},\, \Phi_{\mathrm{sign}},\, \Phi_{\mathrm{rank}},\, \Phi_{\mathrm{prior}},\, \Phi_{\mathrm{causal}}\big), \tag{43}$$

collecting the identity residual, conservation residuals, sign violations, the rank statistic $\hat{\rho}$, rate-prior deviations, and causal/temporal consistency [30, 12]. An authenticity classifier $\mathcal{A}(D) \in \{\mathrm{real}, \mathrm{fake}\}$ trained on Φ deliberately operates *outside* the invariance class of any predictive model over raw buckets, in accordance with the no-go structure of [2]: prediction and authenticity are fundamentally different problems.

9 Value Health Metrics

Table 5: Value health metrics derived from the theory.

Metric	Definition	Reading
Value momentum	$M = C - D = \nu_{E_I} + \nu_{E_D}$	net value creation rate
Relative momentum	$m = M/\mathbb{E}[E]$	growth rate of expected equity
Appreciation ratio	$\kappa_A = \nu_{A_I}/\lvert\nu_{A_D}\rvert$	asset-side asymmetry
Accretion ratio	$\kappa_L = \nu_{L_I}/\lvert\nu_{L_D}\rvert$	liability-side drag
Erosion time	$t^* = E(0)/(-M)$, $M < 0$	expected solvency horizon
Trajectory risk	$\omega^2 = v^\top \Sigma^{(4)} v$	variance of dE
Rank deficiency	ρ of eq. (42)	authenticity signal

10 Numerical Illustration

Table 6: Primitive velocities at $t = 0$ (currency units per year).

(a) Every row satisfies Drift Balance (Theorem 9).

Firm	ν_{A_I}	ν_{A_D}	ν_{L_I}	ν_{L_D}	ν_{E_I}	ν_{E_D}	M	Reading
Expansion	+50.0	−6.0	+16.0	−12.0	+62.0	−22.0	+40.0	creating
Steady	+15.0	−12.5	+10.0	−10.0	+25.0	−22.5	+2.5	neutral
Illustrative	+32.0	−10.0	+18.0	−10.0	+42.0	−28.0	+14.0	creating
Erosion	+4.5	−35.0	+27.0	−1.0	+5.5	−62.0	−56.5	eroding

Table 7: Illustrative firm with $\alpha = 8\%, \beta = 5\%, \gamma = 6\%, \delta = 10\%$ and initial state A_I=400, A_C=100, A_D=200, L_I=300, L_C=100, L_D=100, E_I=150, E_C=10, E_D=40.

(a) All rows satisfy $b^\top X = 0$ to rounding. The E_D tranche falls from 40 to 11.7: destruction of $9.8 + 18.6 = 28.4$ per year is consuming the erosion tranche.

Bucket	$t = 0$	$\mathbb{E}[\,\cdot\,](t = 1)$
A_I / A_C / A_D	400 / 100 / 200	433.3 / 100.0 / 190.2
A	700	723.6
L_I / L_C / L_D	300 / 100 / 100	318.6 / 100.0 / 90.5
L	500	509.0
E_I / E_C / E_D	150 / 10 / 40	192.8 / 10.0 / 11.7
E	200	214.5

Table 8: Balance-sheet plausibility tests.

Test	Authentic ledger	Fabricated ledger	Divergence
Pointwise identity $b^\top X = 0$	Satisfied	Satisfied	0
Flow conservation (Thm. 8)	Satisfied	Violated	High
Drift balance (Thm. 9)	Satisfied	Violated	High
Sign structure (Axiom 3)	Consistent	Contradictory	High
Covariance rank ≤ 8	Deficient (≤ 6)	Full rank	High
Institutional rate priors	Within bounds	Out of bounds	Moderate

(a) A fabricated ledger satisfies the pointwise identity but fails the motion.

For the erosion firm of Table 6a with $E(0) = 200$, Proposition 11 gives $t^* = 200/56.5 \approx 3.54$ years; for the illustrative firm, $M = 14$ per year reproduces the equity growth $200 \to 214.5$ of Table 7a.

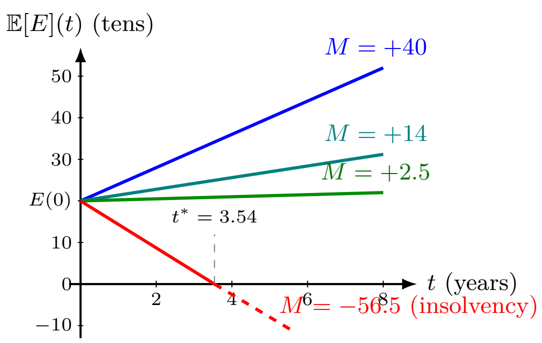

Figure 3: Expected equity trajectories under constant velocities (Proposition 11). The vertical axis is measured in tens of currency units, so $E(0) = 200$ appears at 20; the eroding firm crosses zero at $t^* = E(0)/(-M) = 200/56.5 \approx 3.54$ years.

11 Economic Interpretation

Under pure historical cost, nearly everything sits in A_C until disposal: the classification collapses and the balance sheet hides its own dynamics. Fair-value measurement [7] is precisely what makes the drifts $\alpha, \beta, \gamma, \delta$ observable, and therefore what makes Value-Based Accounting possible. Impairment accounting is a forced reclassification $I \to D$; the Flow Conservation Law then forces the equity narrative to absorb the loss through E_D. The canonical attribution is a stochastic generalization of the clean-surplus relation [5]: equity changes only through value flows, never by fiat. Value momentum M is economic earnings in motion, and $\nu_{E_I} + \nu_{E_D}$ decomposes it into creation and destruction, giving earnings quality a sign-resolved structure [6, 3, 4]. Heavy-tailed return regimes [29] caution that the Gaussian benchmarks of eq. (27) are conservative: the conservation laws survive, the variance formulas do not.

12 Conclusion

We constructed a stochastic theory of the accounting identity and proved:

- a canonical, unique attribution of equity to value-trajectory classes (Theorem 7);
- a pathwise Flow Conservation Law (Theorem 8);
- a Drift Balance law with value momentum $M = C - D$ (Theorem 9);
- covariance singularity of the value state: rank ≤ 8, and ≤ 6 (≤ 4 under strict constancy) canonically (Theorem 14);
- gauge-invariance of the identity and gauge-dependence of the classification (Theorem 17);
- falsification obstructions for fabricated ledgers (Tables 3 and 8).

Ultimately, the accounting identity does not describe a static equality. It describes a conservation law, and the nine buckets describe how value flows through its charge sectors.

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Glossary

Balance Functional
The linear form $b^\top X$ whose vanishing expresses $A = L + E$.

Canonical Attribution
The unique equity decomposition $dE_I = dA_I - dL_D$, $dE_D = dA_D - dL_I$, $dE_C = dA_C - dL_C$ (Theorem 7).

Charge Sectors
The nine buckets, between which the conserved quantity $b^\top X$ is redistributed by transactions but never created or destroyed.

Conservation Law (First Law)
The pathwise identity $dA_I + dA_D = dL_I + dL_D + dE_I + dE_D$ (Theorem 8).

Covariance Singularity
The necessary rank deficiency of Σ_X induced by the balance functional (Theorem 14).

Drift Balance
Equality of expected value velocities across the two sides of the identity (Theorem 9).

Erosion Time
Expected time to insolvency, $t^* = E(0)/(-M)$.

Gauge-Covariant Classification
The frame-dependence of the $I/C/D$ partition under measure or numéraire change (Theorem 17).

Primitive Flows
Appreciation α, depreciation β, accretion γ, amortization δ.

Rank-Deficiency Statistic
$\hat{\rho} = (\lambda_7 + \lambda_8 + \lambda_9)/\sum_i \lambda_i$, an authenticity signal (eq. 42).

Stoichiometric Reading
The interpretation of the identity as a left-null-space conservation constraint of a transaction reaction network.

Trajectory Class
Increasing, constant, or decreasing membership defined by drift sign (Definition 1).

Transaction Operator
A map T with $b^\top T = b^\top$ (eq. 28).

Value Hyperplane
The 8-dimensional slice $\mathcal{V} = \{x : b^\top x = 0\}$ inhabited by every admissible value state.

Value Momentum
$M = C - D$, the net rate of value creation (Definition 10).

Weak versus Strict Constancy
Zero-drift (martingale) versus pathwise-constant readings of the constant class (Remark 4).

The End